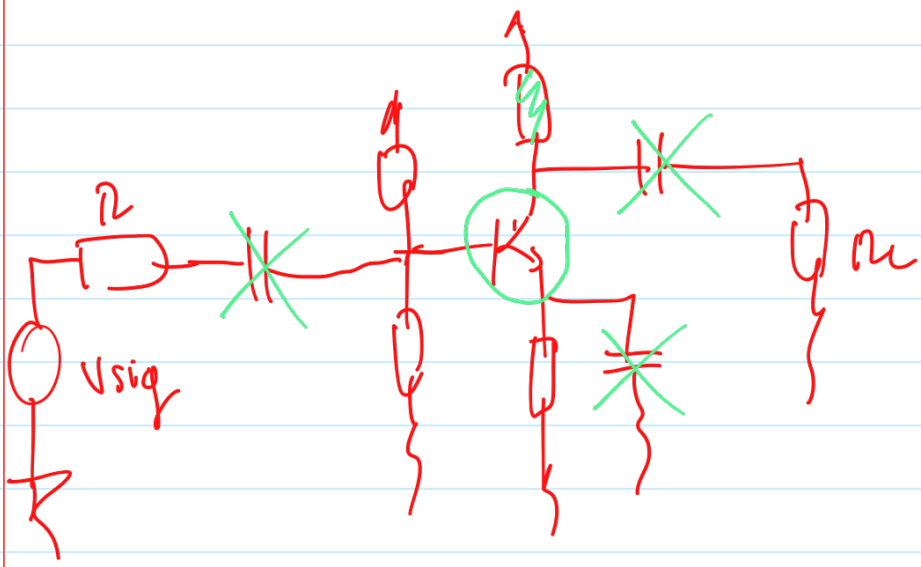
Reading

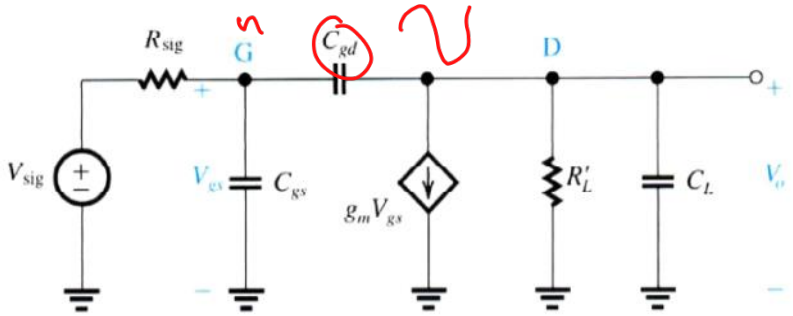
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.7 Other wideband Amplifier Configurations
- Chapter 10.8 Low Frequency Response of Discreate-Circuit CS and CE Amplifiers



Recap

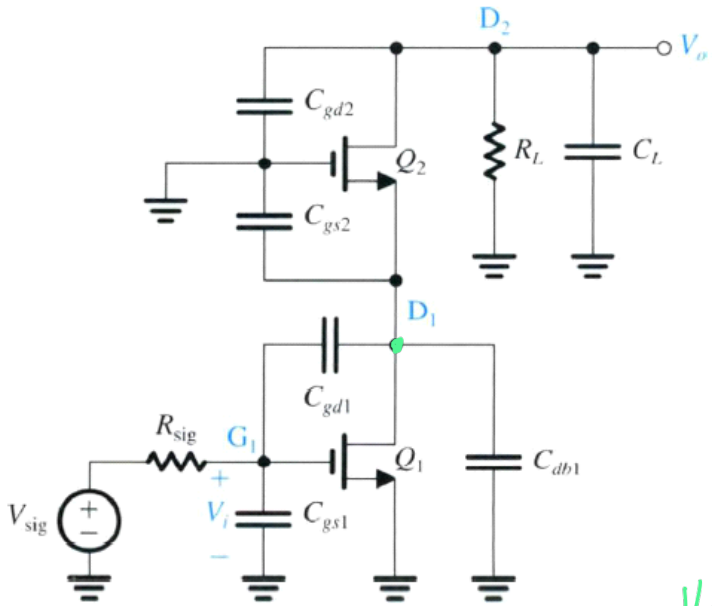
Common Source Amplifier



Cgd Rsig (1+gmRL')

$$\tau_H = C_{gs}R_{gs} + C_{gd}R_{gd} + C_L R_{C_L}$$
$$= C_{gs}R_{sig} + C_{gd}[R_{sig}(1 + g_m R'_L) + R'_L] + C_L R'_L$$

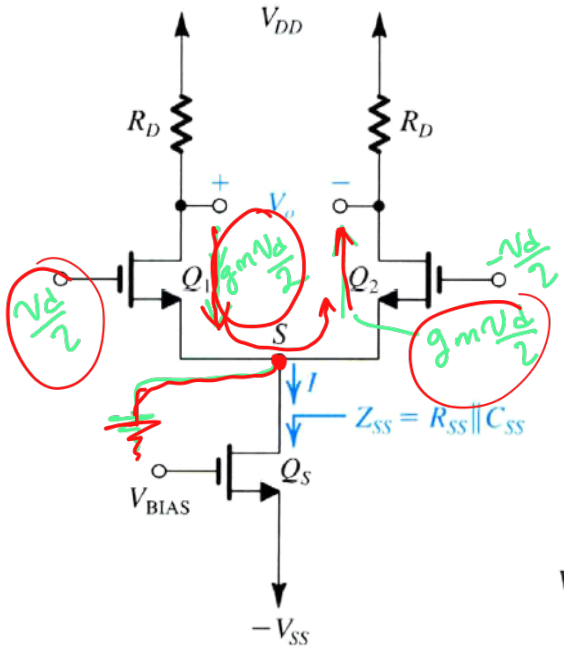
Cascode Amplifier

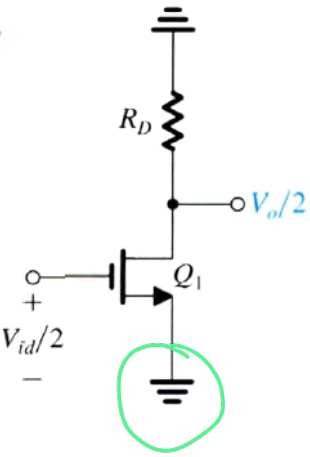


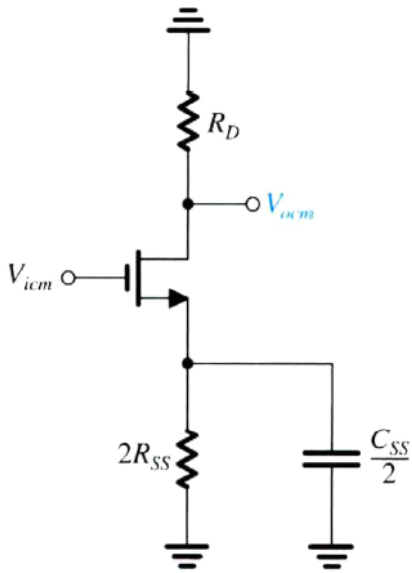
1/gm

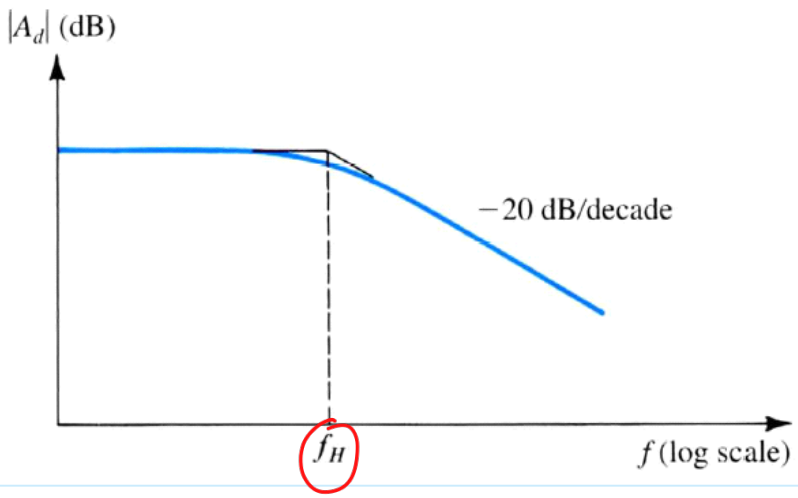
$$\tau_H = C_{gs1}R_{sig} + C_{gd1}\left[(1 + g_{m1}R_{d1})R_{sig} + R_{d1}\right]$$
$$+ (C_{db1} + C_{gs2})R_{d1} + (C_L + C_{gd2})(R_L \| R_o)$$

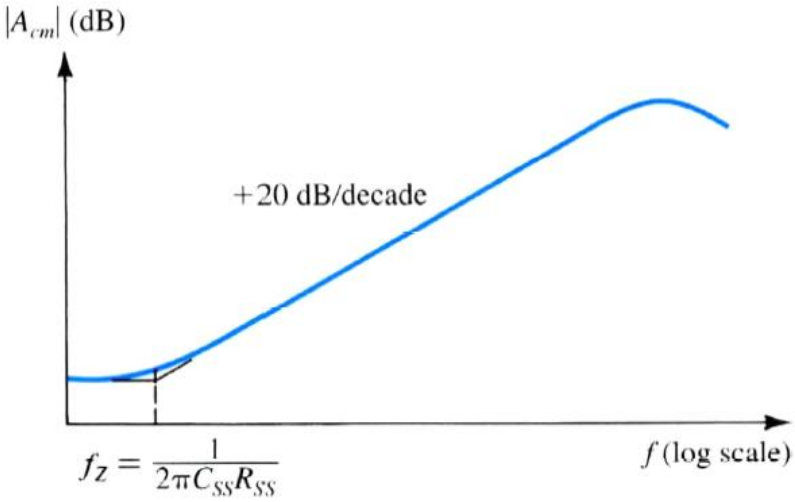
Recap


V_{DD}
R_D
R_D
V_o
Q_1
Q_2
S
I
$Z_{SS} = R_{SS} \| C_{SS}$
Q_S
V_{BIAS}
$-V_{SS}$

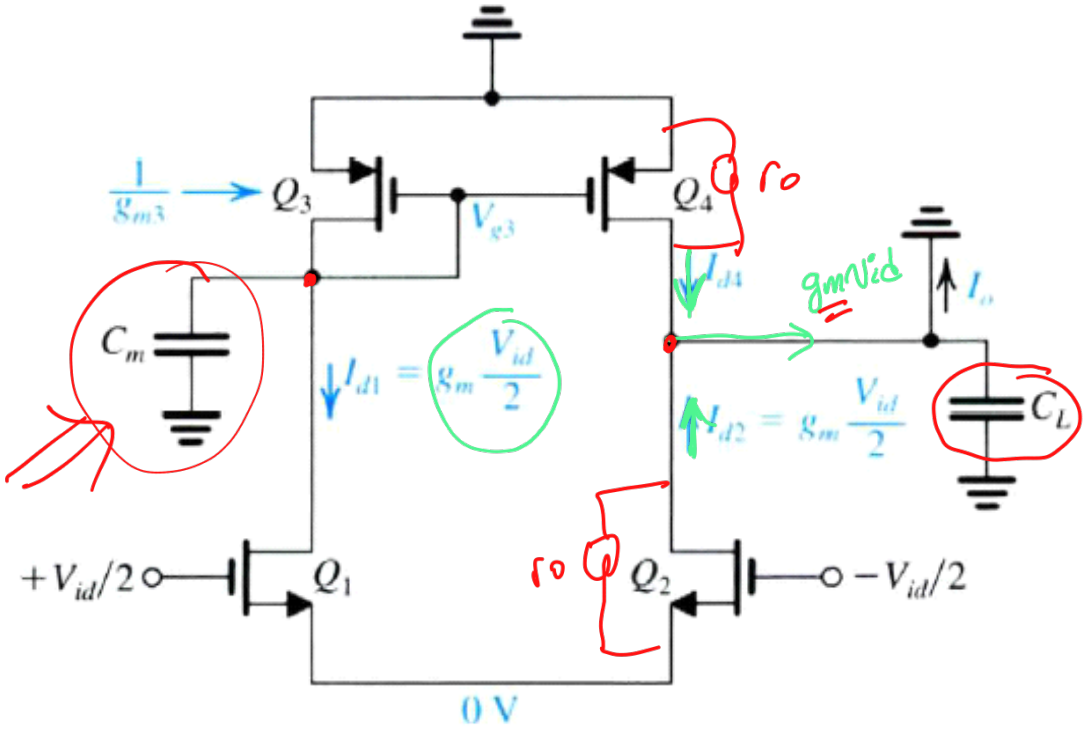

R_D
$V_o/2$
Q_1
$V_{id}/2$

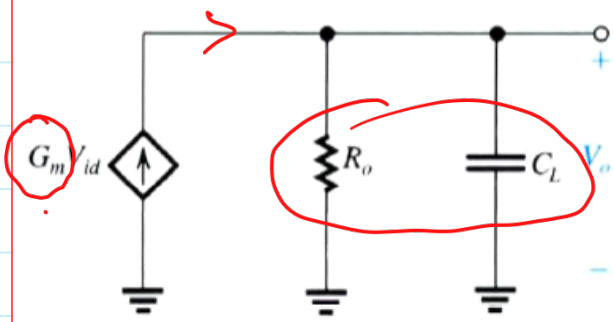

R_D
V_{ocm}
V_{icm}
$2R_{SS}$
$\frac{C_{SS}}{2}$


$|A_d|$ (dB)
−20 dB/decade
f_H
f (log scale)


$|A_{cm}|$ (dB)
+20 dB/decade
$f_z = \frac{1}{2\pi C_{SS} R_{SS}}$
f (log scale)

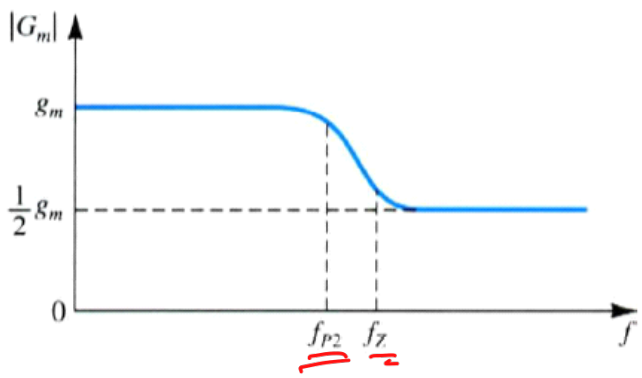
Recap





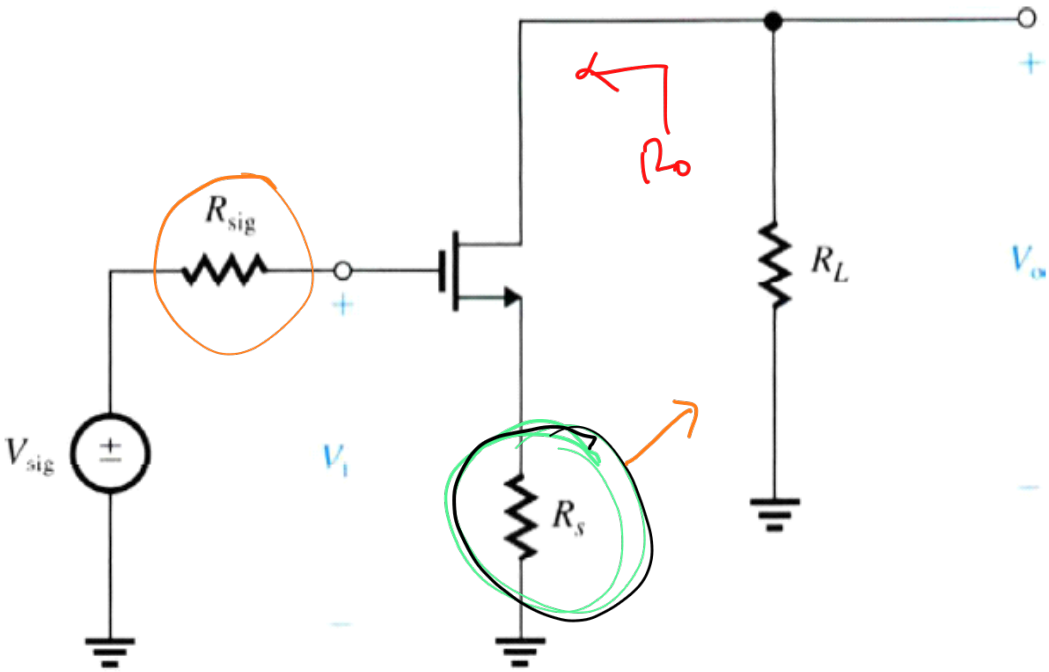
$$f_{P1} = \frac{1}{2\pi C_L R_o}$$

$$G_m \equiv \frac{I_o}{V_{id}} = g_m \frac{1 + s\dfrac{C_m}{2g_{m3}}}{1 + s\dfrac{C_m}{g_{m3}}}$$



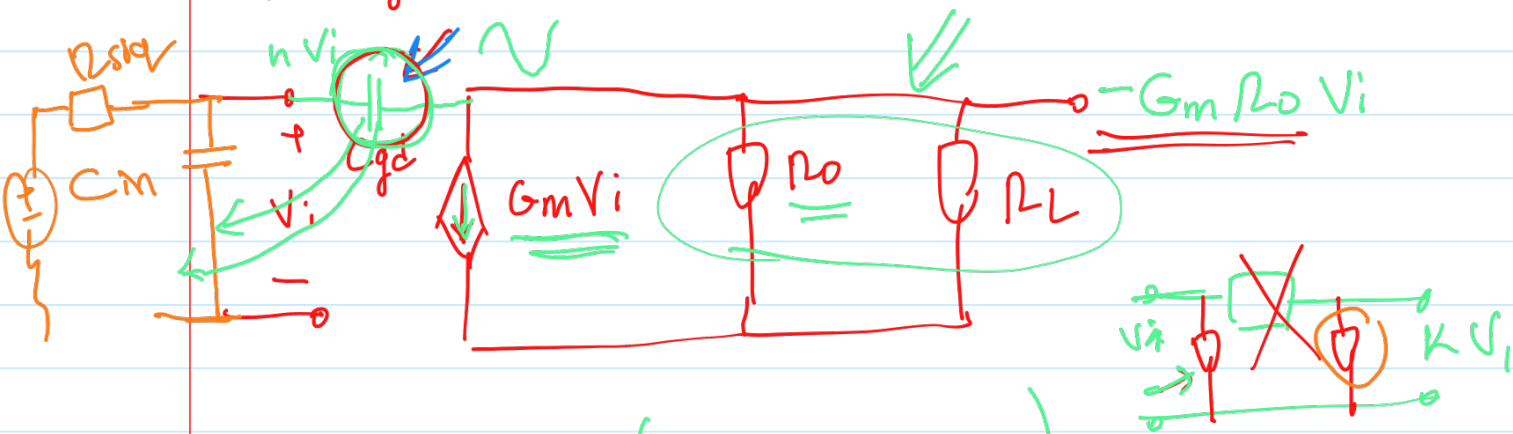
$$\frac{V_o}{V_{id}} = (g_m R_o)\left(\frac{1 + s\dfrac{C_m}{2g_{m3}}}{1 + s\dfrac{C_m}{g_{m3}}}\right)\left(\frac{1}{1 + sC_L R_o}\right)$$

Wideband Amp. by Source/Emitter Degeneration



$$G_m \cong \frac{g_m}{1+g_m R_s}$$

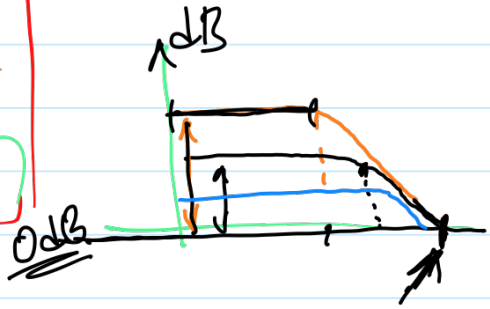
$$R_o = r_o\left(1+g_m R_s\right)$$



$$C_{in} = C_{gd}\left(1+G_m R_o /\!/ R_L\right)$$

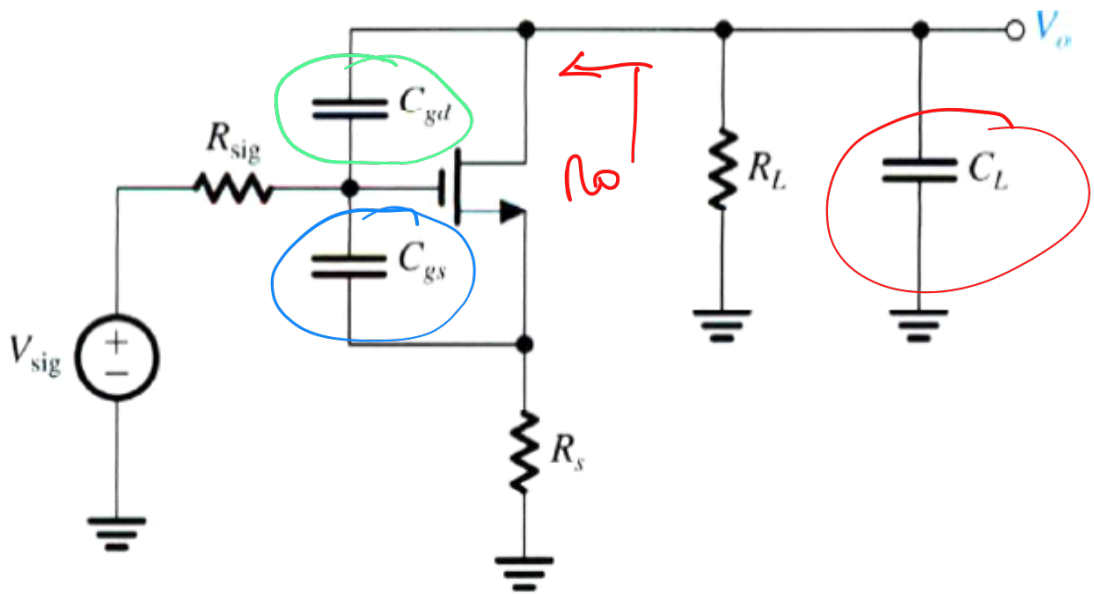
$$= C_{gd}\left(1+|A_M|\right) \cong |A_M| C_{gd}$$

$$f = \frac{1}{2\pi R_{sig} |A_M| C_{gd}}$$



$$GB = |A_M| f = \frac{1}{2\pi R_{sig} C_{gd}}$$

Wideband Amp. by Source/Emitter Degeneration



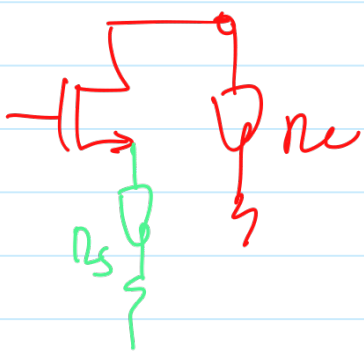
$$R_{gd} = R_{sig}\left(1 + G_m R_L'\right) + R_L'$$

$$R_{gs} = \frac{R_{sig} + R_s + R_{sig} R_s / (r_o + R_L)}{1 + g_m R_s \left(\frac{r_o}{r_o + R_L}\right)}$$

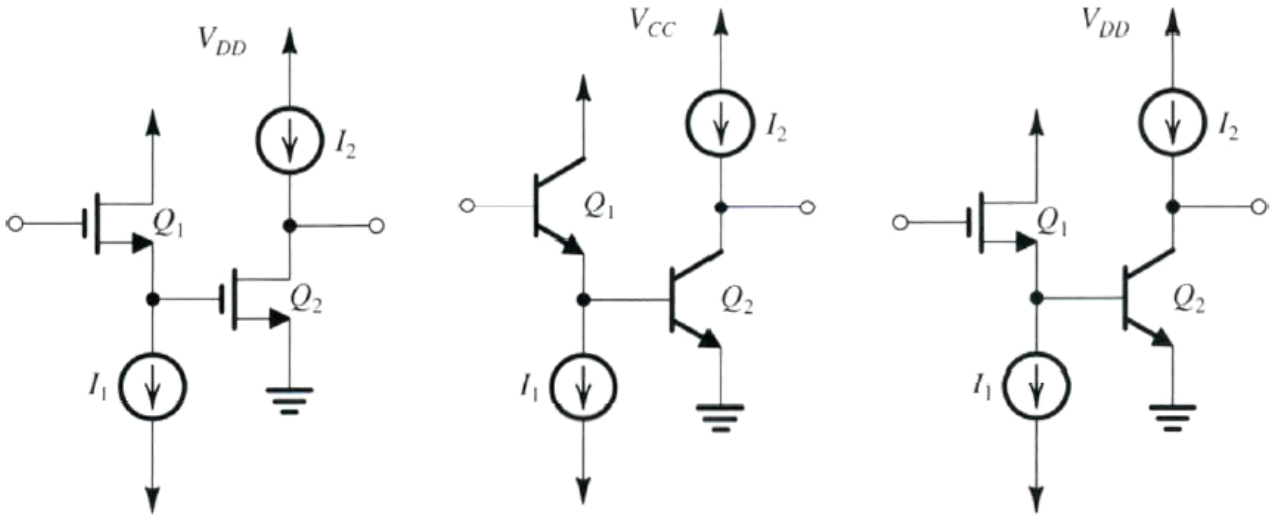
$$R_L = R_L \,/\!/\, R_o = R_L'$$

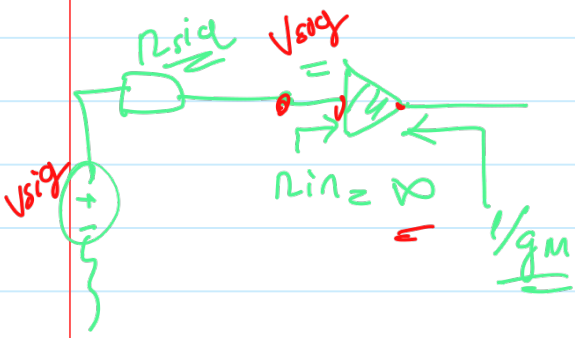
$$\tau_H = C_{gs} R_{gs} + C_{gd} R_{gd} + R_L \cdot C_L$$

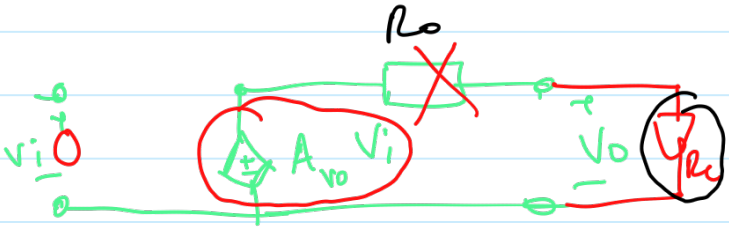
$$f = \frac{1}{2\pi \tau_H}$$

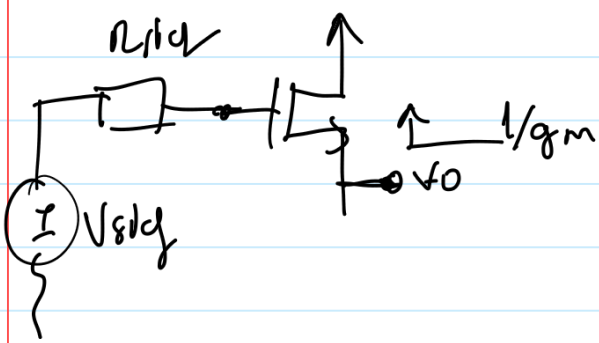


Wideband Amp. by Buffering the Input Signal

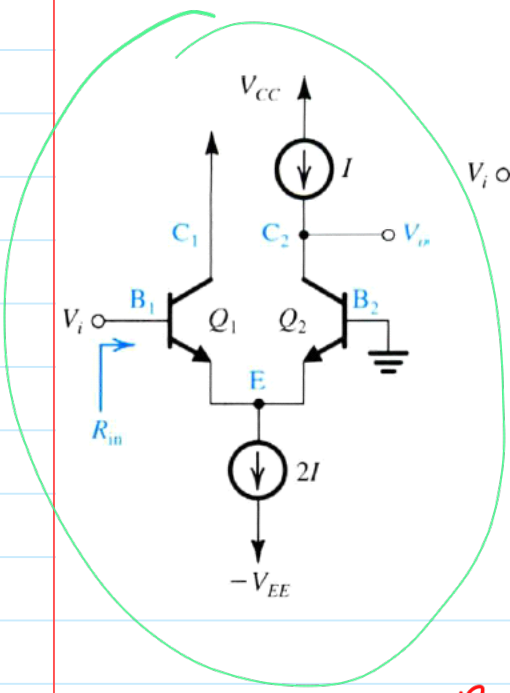


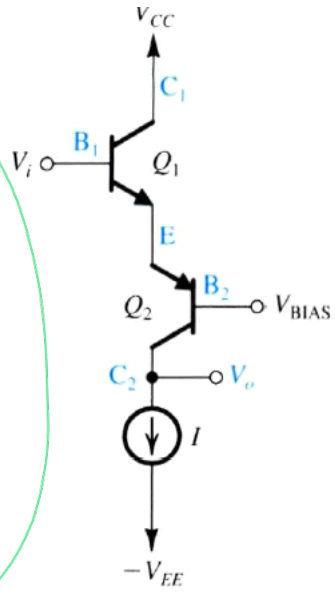


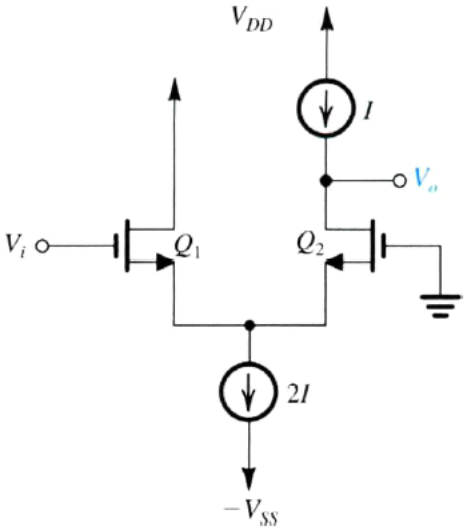


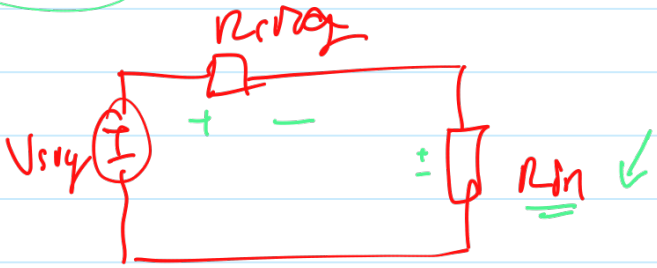


Wideband Amp. by CG/CB with an Input Buffer

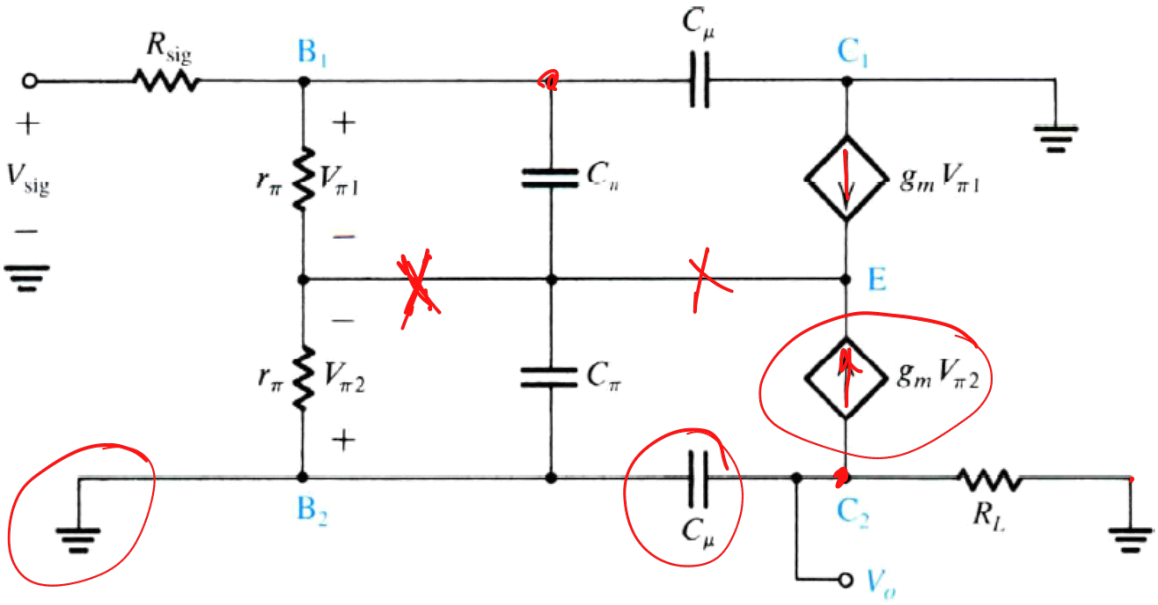


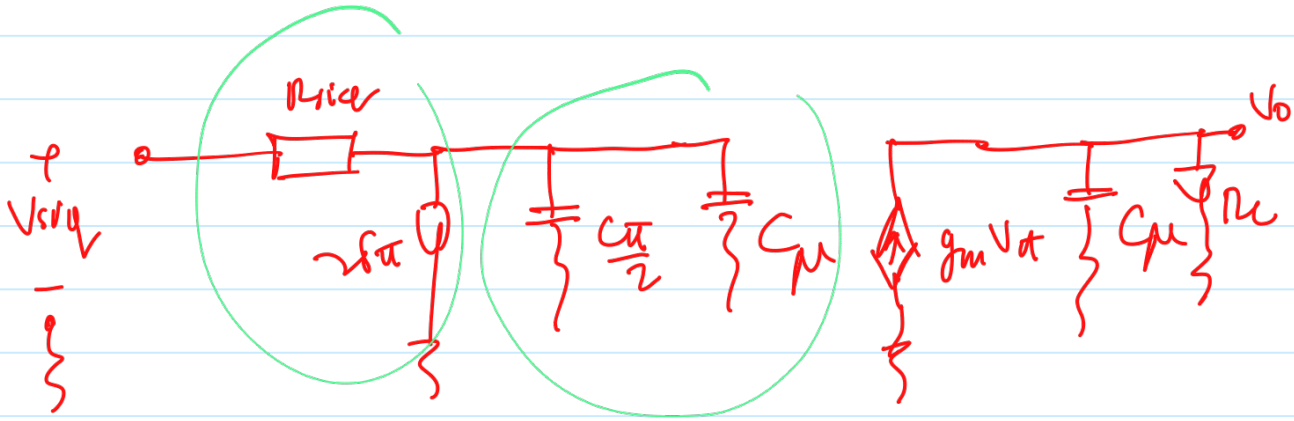






Wideband Amp. by CG/CB with an Input Buffer



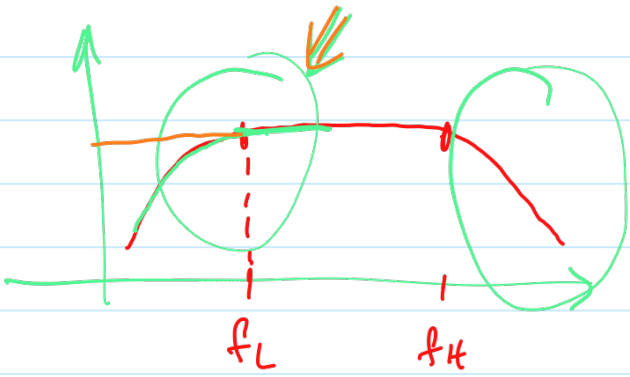


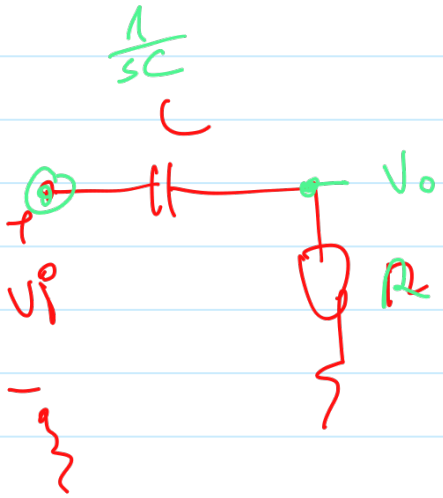
$$f_{P1} = \frac{1}{2\pi\left(\frac{C_\pi}{2} + C_\mu\right)\left(R_{sig} \parallel 2r_\pi\right)}$$

$$f_{P2} = \frac{1}{2\pi C_\mu R_L}$$

$$f_H = \frac{1}{\sqrt{\left(\frac{1}{f_{P1}}\right)^2 + \left(\frac{1}{f_{P2}}\right)^2}}$$

Frequency Response of the High-Pass STC Circuit





$$\left|\frac{V_o}{V_i}\right| = \frac{s}{s+\omega_p}$$

$s \leftarrow j\omega$

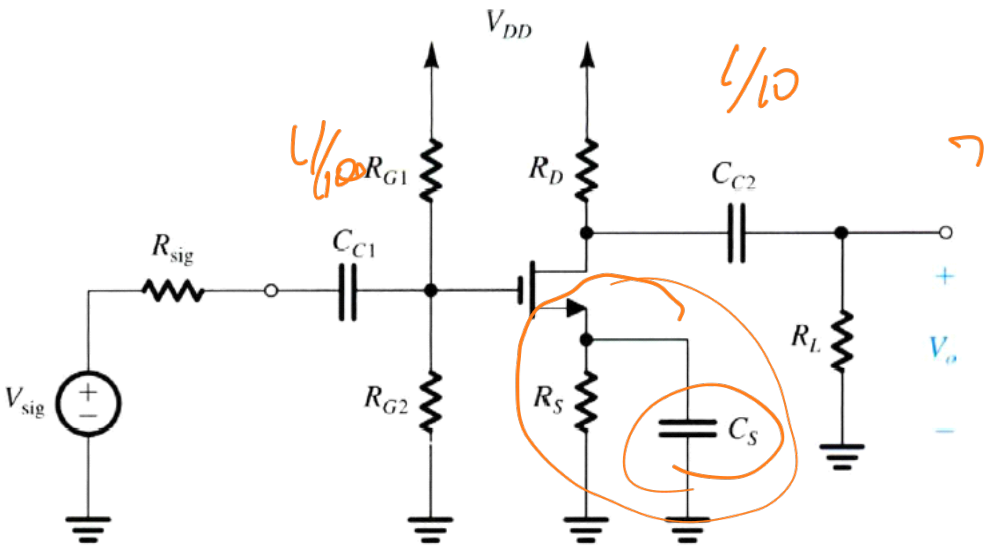
$\omega_p = 1/\tau = \frac{1}{RC}$

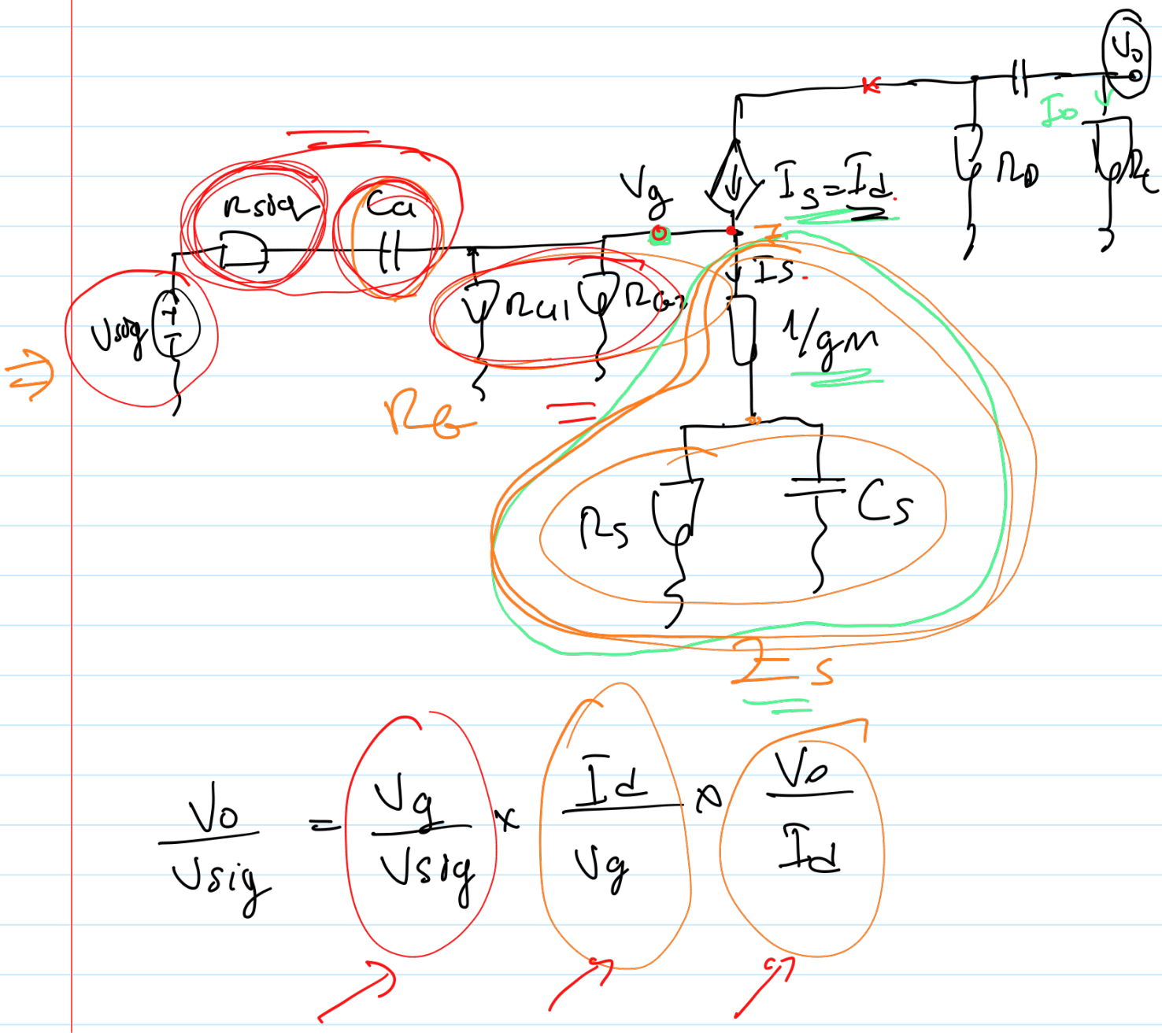
$$\left|\frac{V_o}{V_i}\right| = \frac{1}{\sqrt{1+\left(\frac{\omega_p}{\omega}\right)^2}}$$

$\omega = \omega_p \Rightarrow \left(\frac{\omega_p}{\omega}\right)^2 = 1$, $\sqrt{2}$

$$\left|\frac{V_o}{V_i}\right| = \frac{1}{\sqrt{2}}$$

The CS Amplifier





$$\frac{V_o}{V_{sig}} = \frac{V_g}{V_{sig}} \times \frac{I_d}{V_g} \times \frac{V_o}{I_d}$$

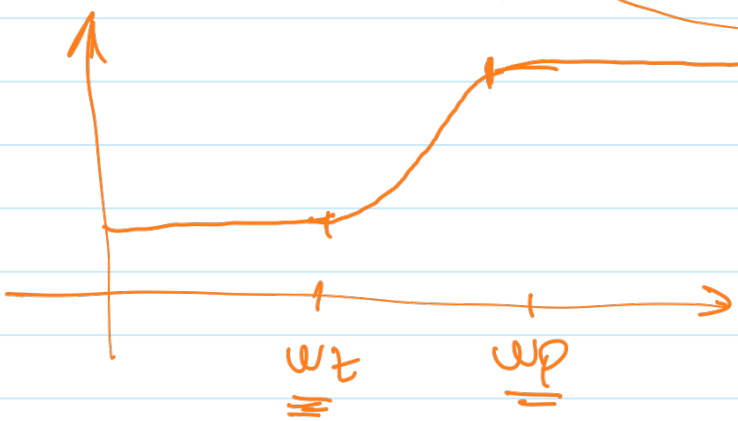
The CS Amplifier

$$\frac{V_g}{V_{sig}} = \frac{R_G}{R_G + R_{sig}} \cdot \frac{s}{s + \frac{1}{C_{C_1}(R_G + R_{sig})}}$$

ω_{p_1}

$$\frac{I_d}{V_g} = g_m \cdot \frac{s + \frac{1}{C_S R_S}}{s + \frac{g_m + 1/R_S}{C_S}}$$

ω_z ω_{p_2}

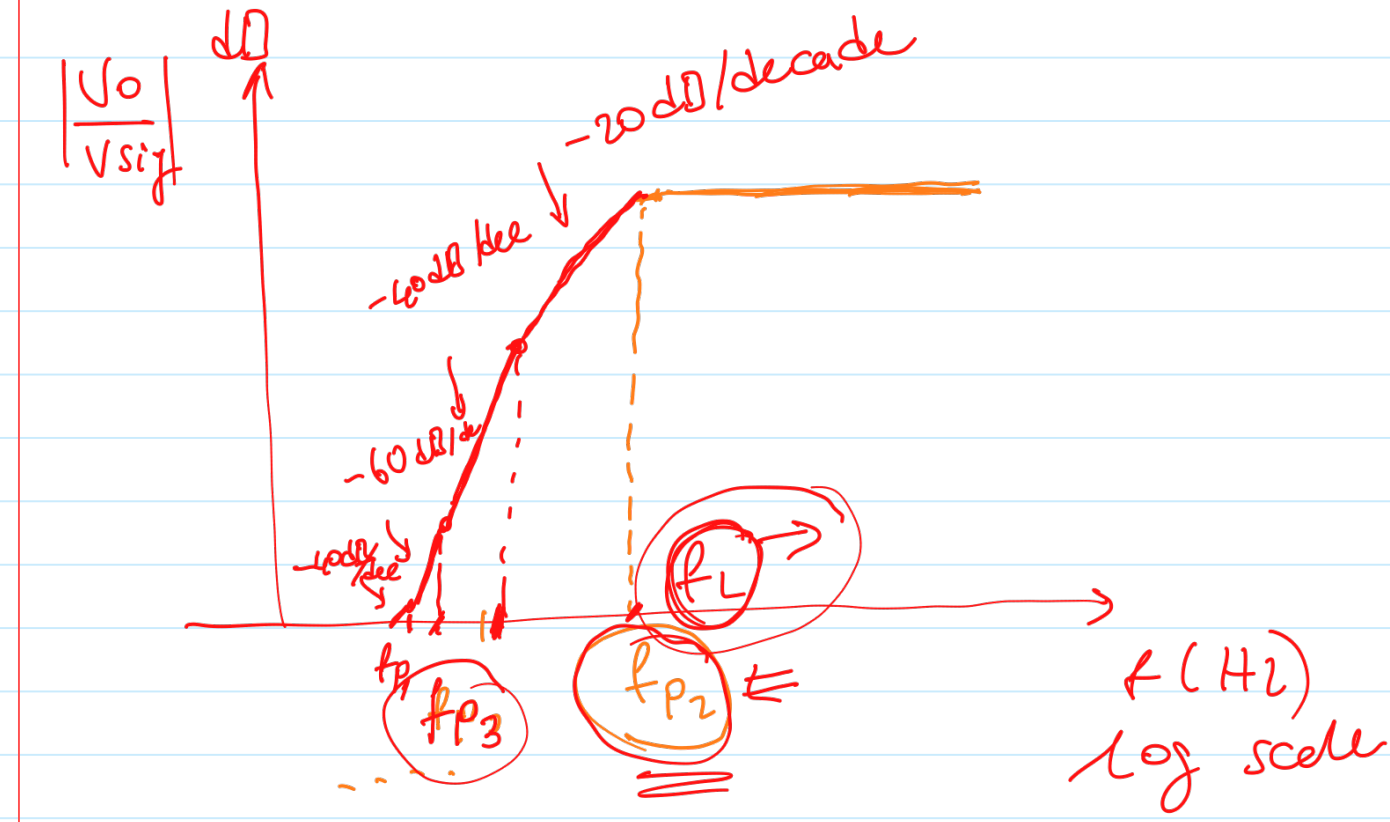


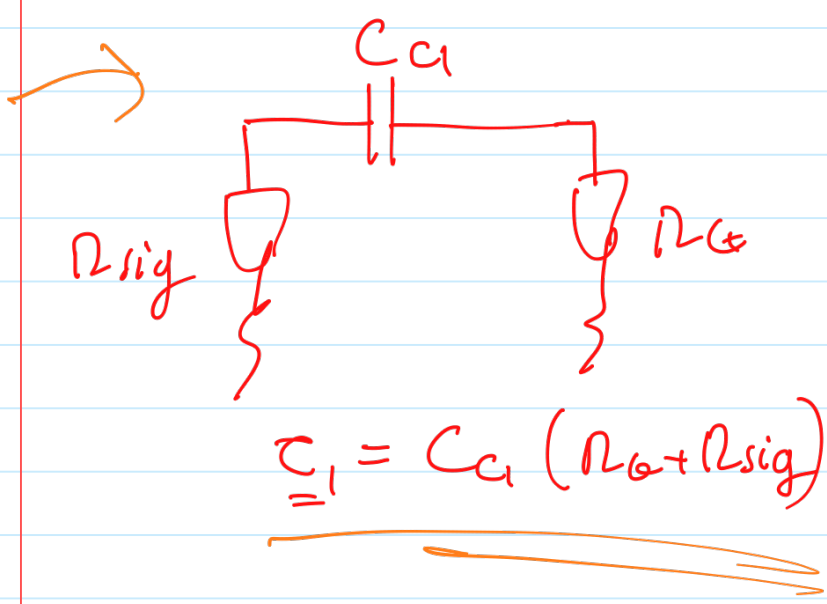
$$\frac{V_o}{I_d} = -\frac{R_D R_L}{R_D + R_L} \cdot \frac{s}{s + \frac{1}{C_{C_2}(R_D + R_L)}}$$

ω_p

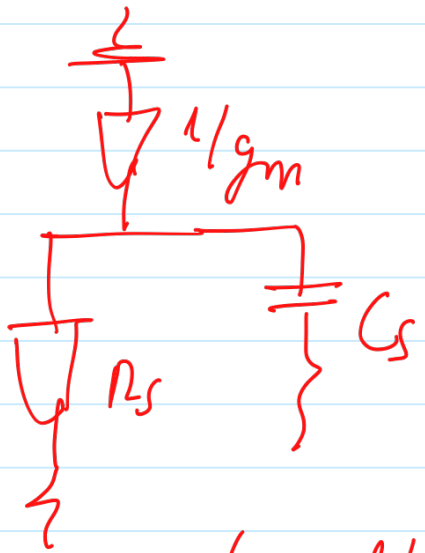
$$f_L \cong \sqrt{f_{p_1}^2 + f_{p_2}^2 + f_p^2 - 2f_z^2}$$

Template

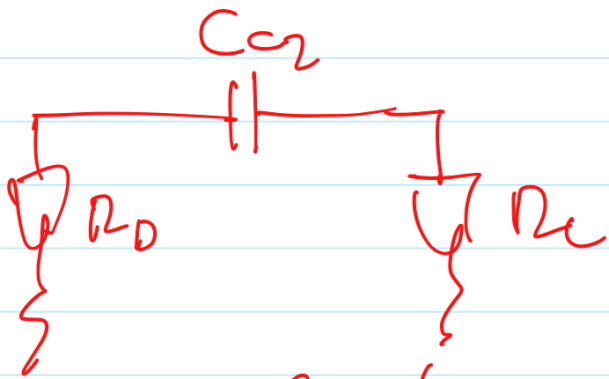




$$\tau_1 = C_{C1}(R_G + R_{sig})$$



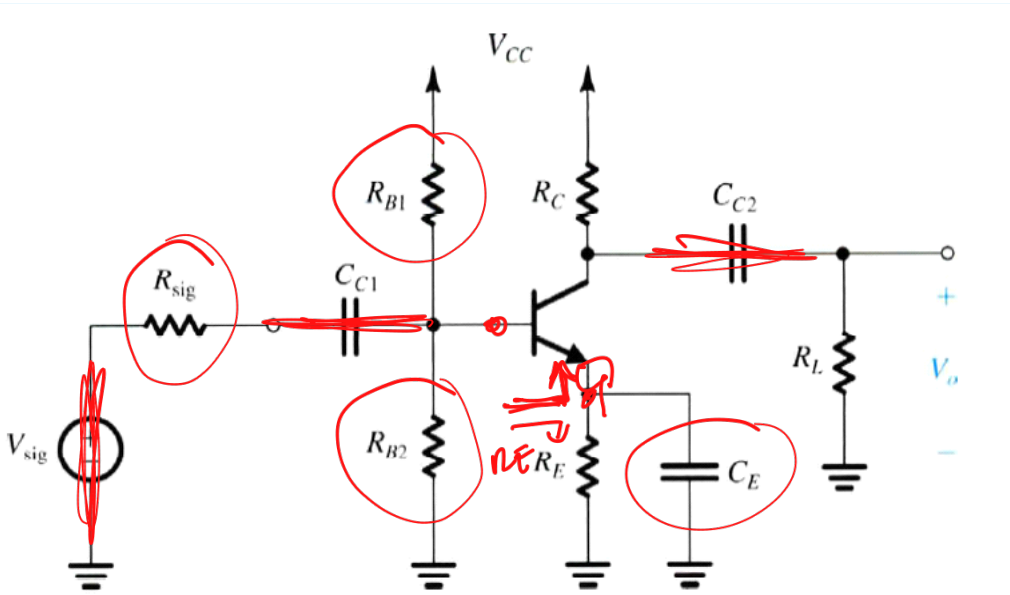
$$\tau_S = C_S(R_S \,//\, 1/g_m)$$



$$\tau_2 = C_{C2}(R_D + R_L)$$

The Method of Short-Circuit Time Constants

The CE Amplifier



$$R_{C1} = R_{sig} + (R_{B1} \,//\, R_{B2} \,//\, r_\pi)$$

$$\tau_{C1} = C_{C1} \cdot R_{C1}$$

$$R_{C2} = (R_C + R_L)$$

$$\tau_{C2} = R_{C2} C_{C2}$$

$$R_{CE} = R_E \,//\left(r_e + \frac{R_{sig} \,//\, R_{B1} \,//\, R_{B2}}{\beta + 1}\right)$$

$$\tau_{CE} = R_{CE} \cdot C_{CE}$$

$$f_L \approx \frac{1}{2\pi} \sum_{i=1}^{n} \frac{1}{\tau_i} = \frac{1}{2\pi}\left(\frac{1}{\tau_{C1}} + \frac{1}{\tau_{C2}} + \frac{1}{\tau_{CE}}\right)$$

Template

